Alex Pawelko

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Education

University College London, London, England — September 2026 — Present
PhD Student in Mathematics

- Supervised by Professor Aleksander Doan.
- Research Interests: Differential geometry and geometric analysis, particularly the study of special geometric structures with a focus on special holonomy metrics and their associated calibrated geometries and gauge theories.

University of Waterloo, Waterloo, Ontario, Canada — September 2021 — April 2026
Bachelor of Mathematics, Honours Pure Mathematics Co-op, Computer Science Minor — 96% Average

- Graduated with Dean's Honours and 13x Term Distinction (every time eligible).

Teaching Experience

Department of Mathematics, University College London — London, England
Teaching Assistant — October 2026 — Present

- Graded homework assignments for courses on manifolds (MATH0019) and advanced linear algebra (MATH0014).

Faculty of Mathematics, University of Waterloo — Waterloo, Ontario, Canada
Teaching Assistant (part-time) — January 2026 — April 2026

- Held office hours for introductory calculus courses (MATH 127, 128, 137, 138, 237).

Faculty of Mathematics, University of Waterloo — Waterloo, Ontario, Canada
Teaching Assistant (part-time) — April 2025 — August 2025

- Held office hours for introductory calculus courses (MATH 127, 128, 137, 138, 237).

Professional Experience

Department of Pure Mathematics, University of Waterloo — Waterloo, Ontario, Canada
Research Assistant (co-op) — September 2025 — December 2025

- Explored differential geometry research project, "Infinite-dimensional analogues of exceptional geometric structures" supervised by Professors Xuemiao Chen and Spiro Karigiannis.
- Funded by Mathematics Undergraduate Research Award from the Faculty of Mathematics.

School of Computer Science, University of Waterloo — Waterloo, Ontario, Canada
Research Fellow (co-op) — January 2025 — March 2025

- Formalized semantics of probabilistic programming languages using categorical logic and measure theory as part of programming language theory research project supervised by Professor Yizhou Zhang.
- Funded by Undergraduate Research Fellowship from the School of Computer Science (ended early due to medical leave).

Department of Pure Mathematics, University of Waterloo — Waterloo, Ontario, Canada
Research Assistant (co-op) — May 2024 — August 2024

- Investigated differential geometry research project, "Aspects of symplectic theory that generalize to G_2 geometry" supervised by Professor Spiro Karigiannis.
- Funded by NSERC Undergraduate Student Research Award.

Tactic Studios — London, Ontario, Canada
Game Programmer (co-op) — September 2023 — December 2023

- Programmed gameplay, user interfaces, graphics, and networking features for Killer Inn published by Square Enix.

Eden Industries — Remote
Game Programmer (co-op) — May 2022 — August 2022 and January 2023 — April 2023

- Programmed graphics, gameplay, and engine features in three upcoming titles.

Podcrash — Remote
Software Development Lead — April 2021 — July 2021

- Led a team of software developers making second-party add-ons for Minecraft.

Podcrash Remote
Software Developer September 2020 — April 2021

- Programmed server-side game logic and user authentication for second-party add-ons for Minecraft.

Beyond Skyrim Remote
Implementation Developer (freelance) July 2017 — August 2020

- Implemented gameplay features in fan game project for Skyrim.

Invited and Contributed Academic Talks

UW Differential Geometry Working Seminar (all contributed)

- Adiabatic Limits of Coassociative Fibrations II, August 12th, 2026
- Wild Speculation on Homotopy Invariants of G_2-structures, August 5th, 2026
- Adiabatic Limits of Coassociative Fibrations I, June 17th, 2026
- Morse Theory via Harmonic Oscillators, April 16th 2026
- Riemannian Geometry of Knot Spaces, December 11th 2025
- Calibrated Geometry of a Strongly Nondegenerate Knot Space, September 18th 2025
- Gerbes of Coassociative Submanifolds and the First Chern Class, August 14th 2025
- The Formal Kähler Structure of the G_2 Knot Space, May 22nd 2025
- Prequantum Line Bundles and Geometric Quantization, November 20th 2024
- Strongly Nondegenerate Forms and their Associated Structures on Higher Knot Spaces, August 7th 2024
- Symmetry Reduction and the Quest for G_2 Moment Maps, May 29th 2024

Canadian Undergraduate Mathematics Conference (all contributed)

- What's up with the 7-dimensional cross product? June 25th, 2026
- How Special is Curvature? June 26th, 2025
- Quantifying the Fundamental Theorem of Calculus, July 9th 2024

UW Pure Math Club Short Attention Span Math Seminars (all contributed)

- Let's Prove the Poincaré Conjecture, March 18th 2026
- Differential Geometers Suck at Naming Things, March 7th 2025
- 2-outta-3 Ain't Bad, November 7th 2024
- The Cross Product in $\mathbb{R}^7$ (and Other Fun Adventures with Normed Division Algebras), March 14th 2024
- Russian Man Approximates Eigenvalues With This One Weird Trick^{tm}, Mathematicians Hate Him!, July 13th 2023

Administrative Service & Other Experience

- University of Waterloo Senator, September 2025 — August 2026
- Vice-President of the Organizing Committee, Canadian Undergraduate Mathematics Conference 2025
- Vice-President, Academic of the Mathematics Society of the University of Waterloo, September 2024 — December 2024
- Taught many midterm and final exam review tutorials for introductory calculus and linear algebra courses at UW (MATH 137, 138, 145, 146, 147, 148, 235, 237, 245, 247), September 2023 — April 2026
- President of the Pure Math Club of the University of Waterloo, January 2025 — April 2025
- Residence Don, Conrad Grebel University College, January 2024 — April 2024
- Service on over 15 committees for the University of Waterloo, the Waterloo Undergraduate Student Association, the Mathematics Society of the University of Waterloo, and Conrad Grebel University College

Awards, Scholarships, and Honours

- K.D. Fryer Gold Medal (as the member of my graduating class "best exemplifying high academic standing and good student citizenship"), University of Waterloo Faculty of Mathematics, October 2026
- John Hawkes Scholarship (selected as a beginning PhD student "on the basis of academic merit and potential"), University College London Mathematics, September 2026
- Honourary Lifetime Member of the Mathematics Society of the University of Waterloo, March 2026
- Mike Vangoch Memorial Scholarship (for outstanding achievement as a fourth year Pure Math student), University of Waterloo, December 2025
- Mathematics Undergraduate Research Award, University of Waterloo Faculty of Mathematics, September 2025
- Honourary Member for Life, Pure Math Club of the University of Waterloo, May 2025
- President's Research Award, University of Waterloo, January 2025
- President's Research Award, University of Waterloo, May 2024
- Natural Sciences and Engineering Research Council of Canada (NSERC) Undergraduate Student Research Award, May 2024
- Conrad Grebel University College Jacob Andres Achievement Award, November 2023
- Conrad Grebel University College Anniversary Legacy Award, March 2023
- Software Engineering Entrance Scholarship, University of Waterloo, September 2021
- President's Scholarship of Distinction, University of Waterloo, September 2021